

Overnight lowering of free fatty acids with Acipimox improves insulin resistance and glucose tolerance in obese diabetic and nondiabetic subjects.

Obesity is commonly associated with elevated plasma free fatty acid (FFA) levels, as well as with insulin resistance and hyperinsulinemia, two important cardiovascular risk factors. What causes insulin resistance and hyperinsulinemia in obesity remains uncertain. Here, we have tested the hypothesis that FFAs are the link between obesity and insulin resistance/hyperinsulinemia and that, therefore, lowering of chronically elevated plasma FFA levels would improve insulin resistance/hyperinsulinemia and glucose tolerance in obese nondiabetic and diabetic subjects. Acipimox (250 mg), a long-acting antilipolytic drug, or placebo was given overnight (at 7:00 P.M., 1:00 A.M., 7:00 A.M.) to 9 lean control subjects, 13 obese nondiabetic subjects, 10 obese subjects with impaired glucose tolerance, and 11 patients with type 2 diabetes. Euglycemic-hyperinsulinemic clamps and oral glucose tolerance tests (75 g) were performed on separate mornings after overnight Acipimox or placebo treatment. In the three obese study groups, Acipimox lowered fasting levels of plasma FFAs (by 60-70%) and plasma insulin (by approximately 50%). Insulin-stimulated glucose uptake during euglycemic-hyperinsulinemic clamping was more than twofold higher after Acipimox than after placebo. Areas under the glucose and insulin curves during oral glucose tolerance testing were both approximately 30% lower after Acipimox administration than after placebo. We conclude that lowering of elevated plasma FFA levels can reduce insulin resistance/hyperinsulinemia and improve oral glucose tolerance in lean and obese nondiabetic subjects and in obese patients with type 2 diabetes.